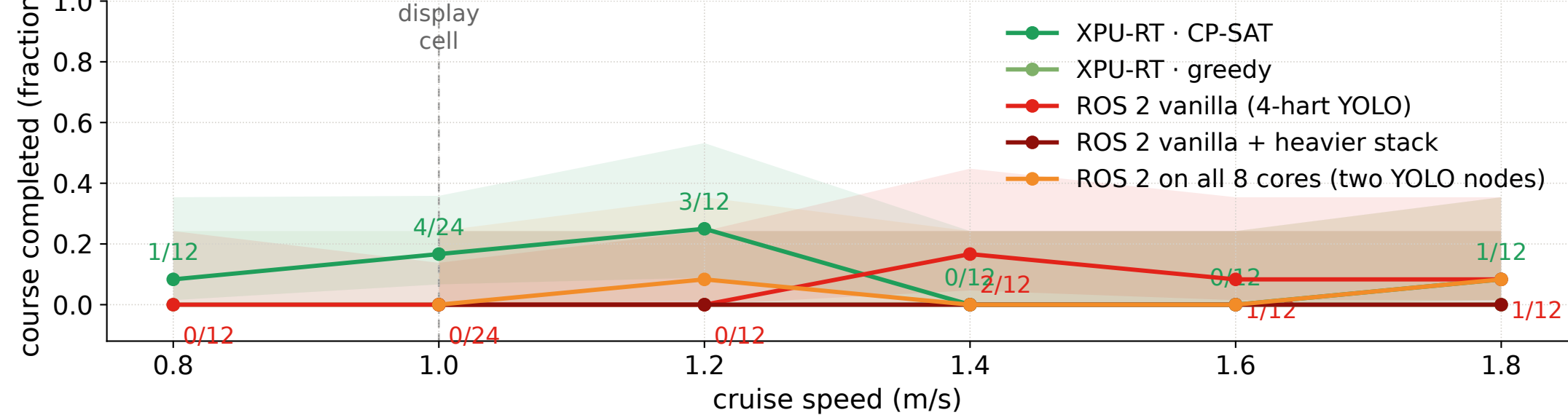
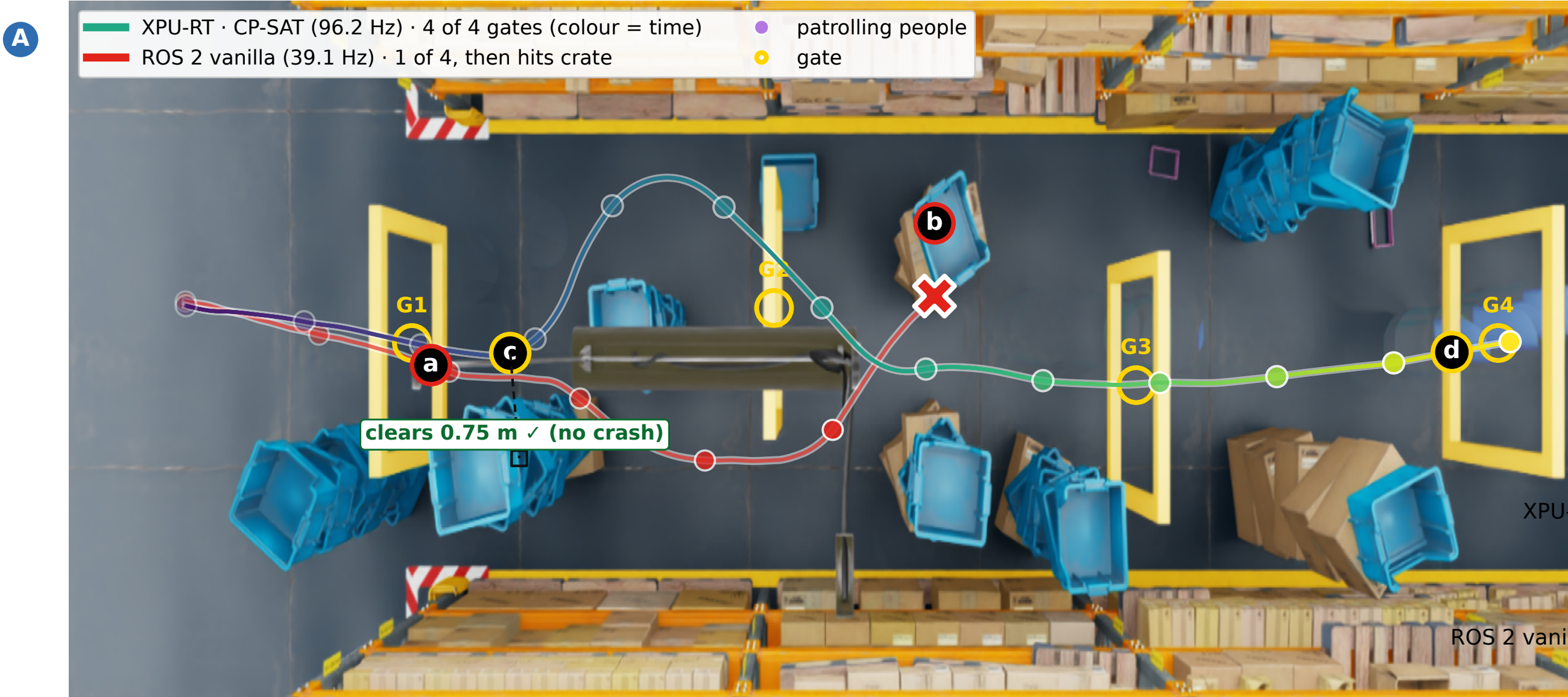
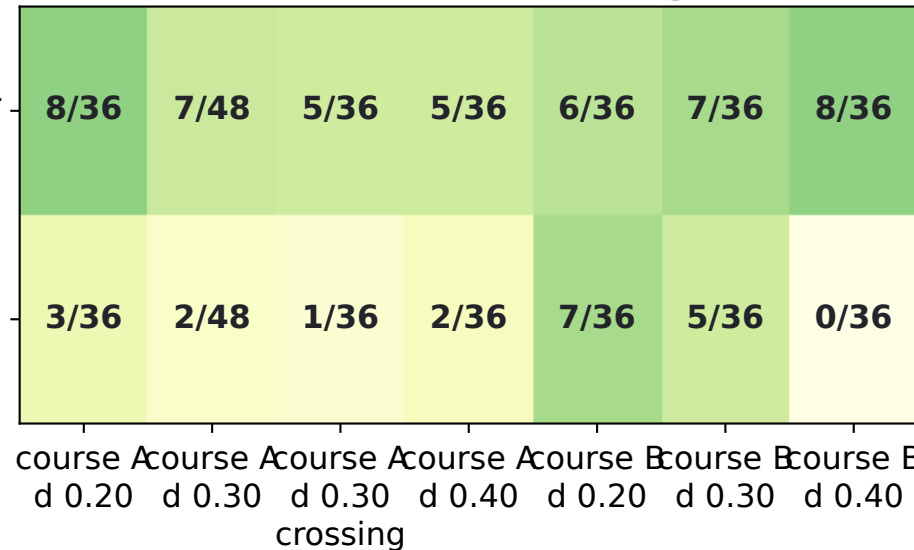


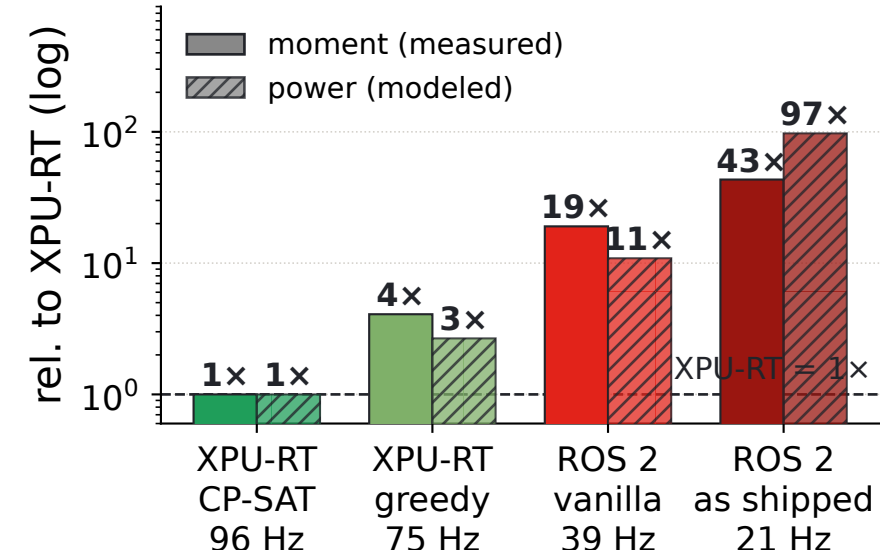
Warehouse gate-course showdown — same scene and controller (1.0 m/s)
only what the K1 delivers differs: control every 10.4 vs 25.5 ms, camera→control latency and goal cadence, replayed
B Flight envelope: success against cruise speed, each arm at its own K1 cadence
only what the K1 delivers differs: control every 10.4 vs 25.5 ms, camera→control latency and goal cadence, replayed



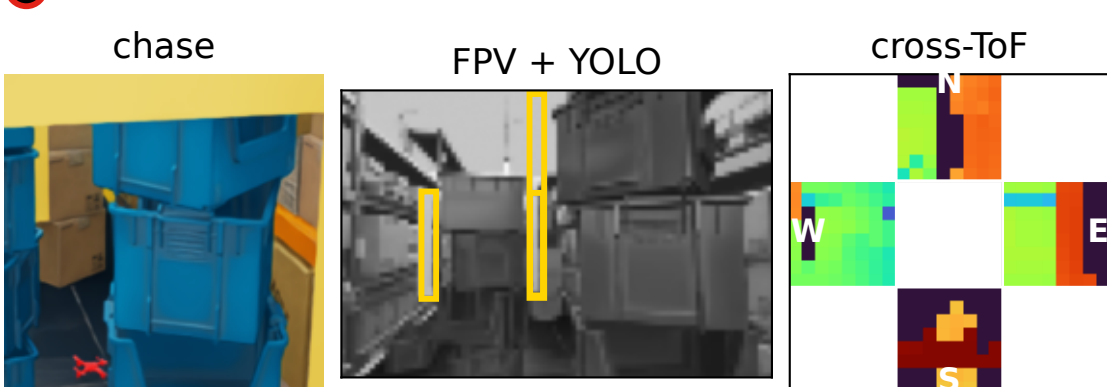
C Generalisation: environments (1.0–1.4 m/s pooled, one flight per seed)



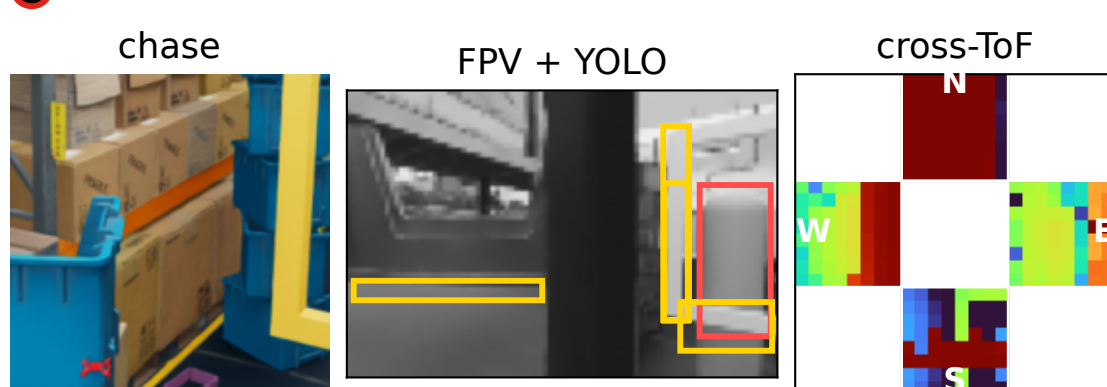
D Mechanism: the baseline thrashes (moment, power)



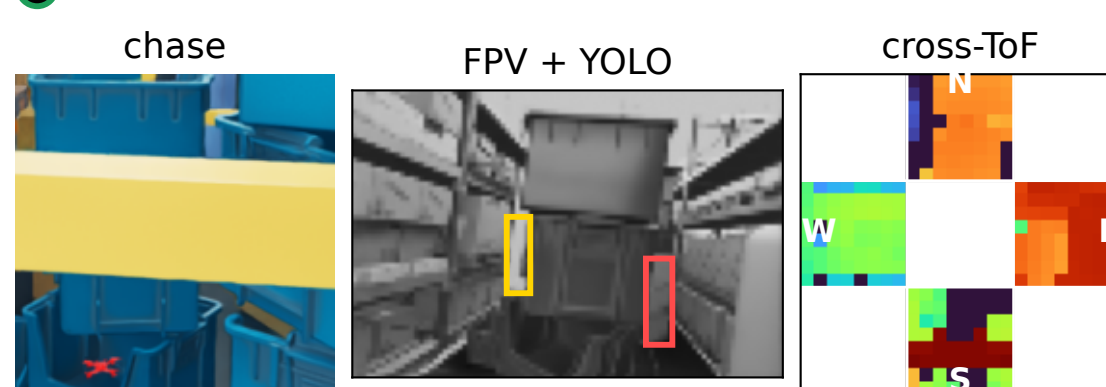
a ROS 2 39 Hz · clears G1 · t=3.5s



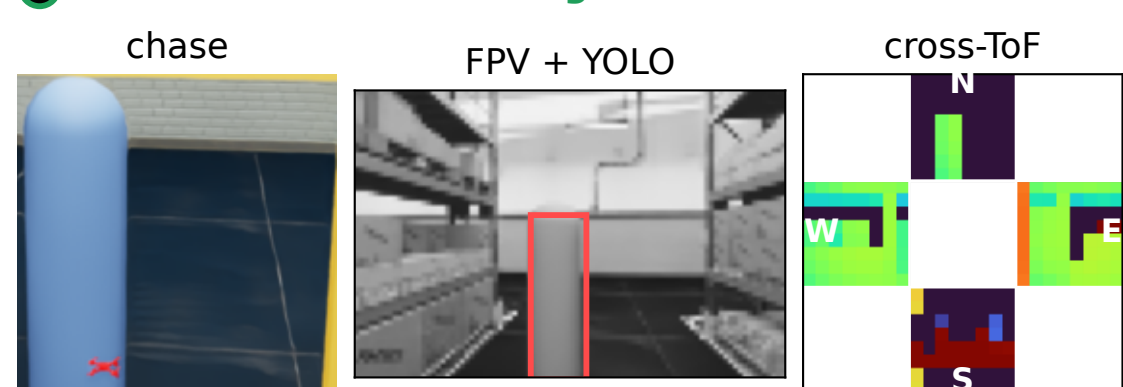
b ROS 2 39 Hz · hits crate after G1 · t=9.5s



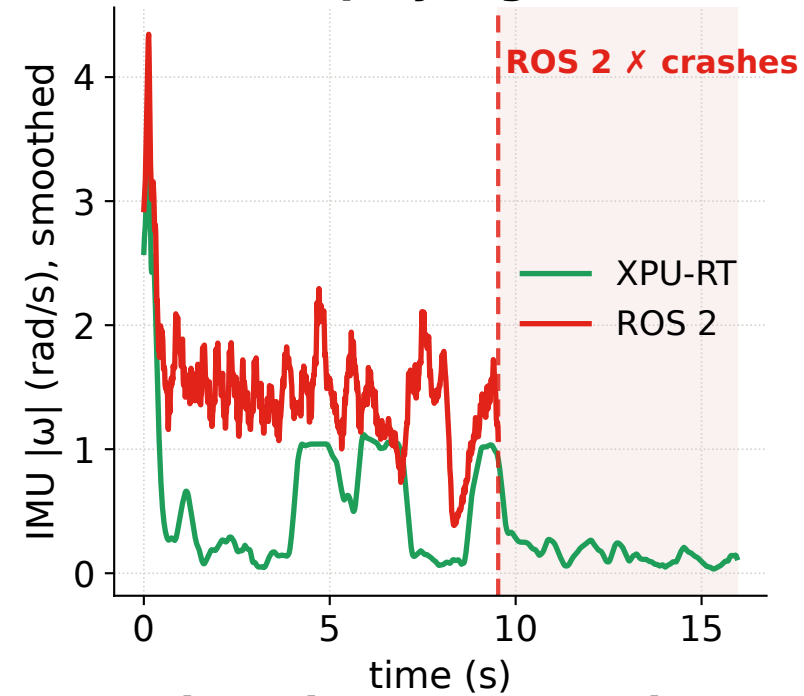
c XPU-RT 96 Hz · clears crate 0.75 m · t=4.3s



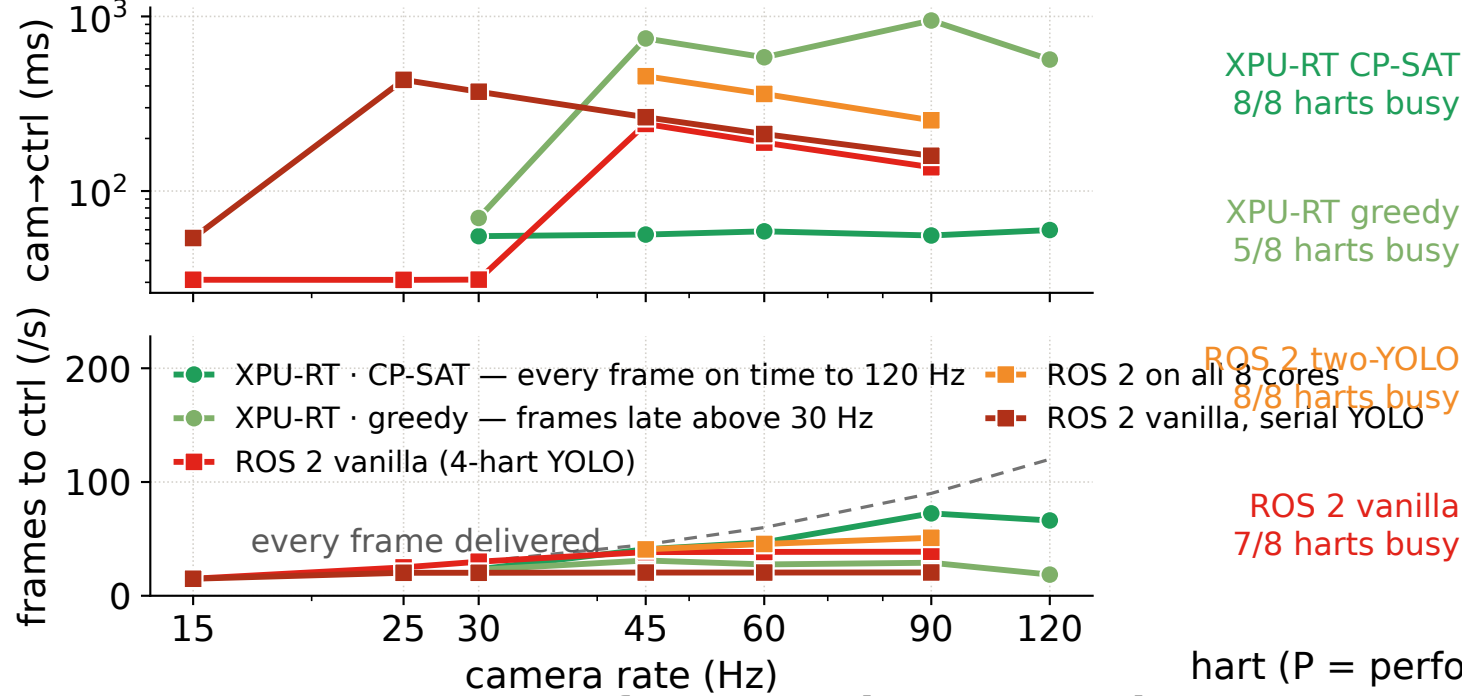
d XPU-RT 96 Hz · reaches gate · t=15.3s



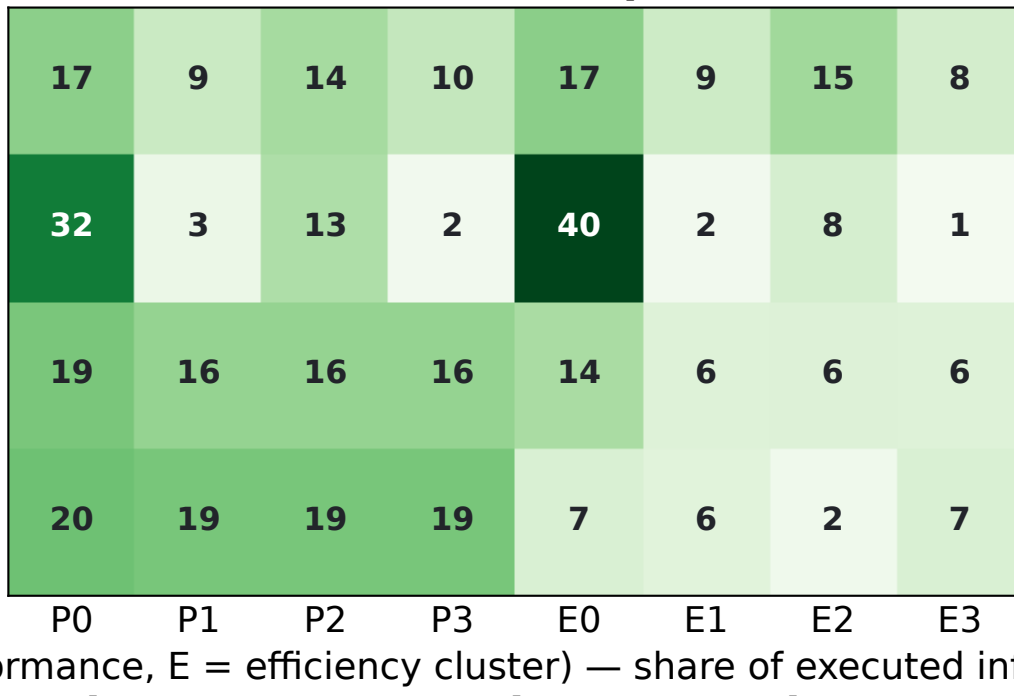
E Body-rate magnitude of the display flights



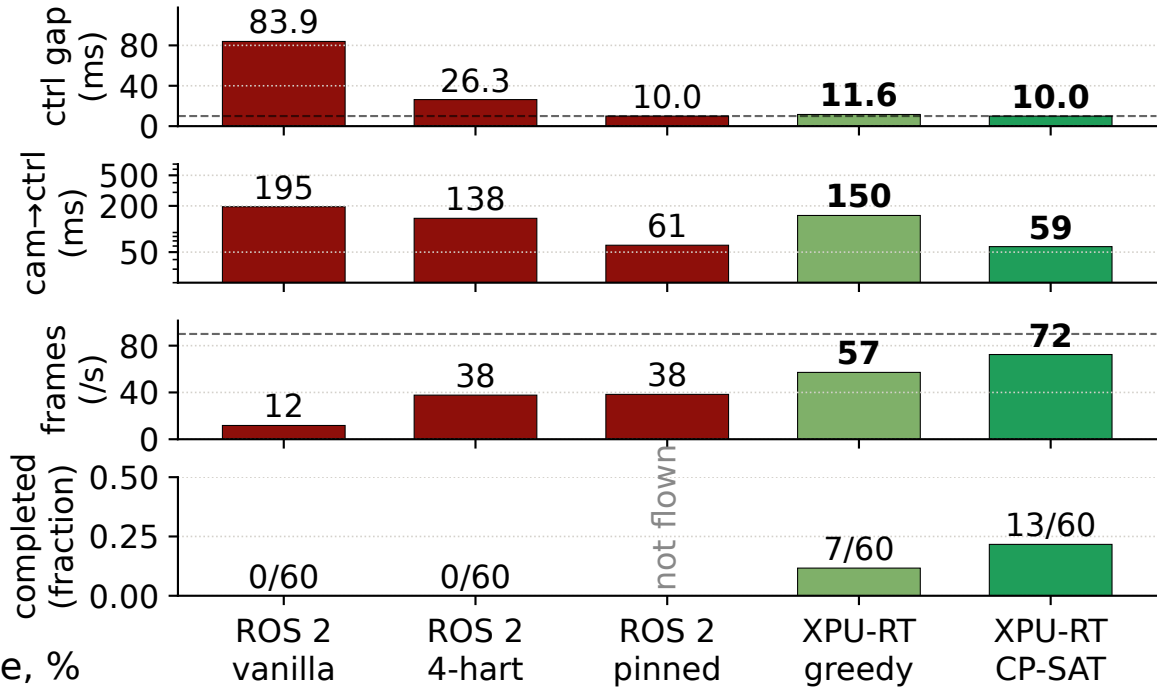
F Camera rate on the K1: camera→control latency and frames delivered, same kernels



G Where the work lands, 45 Hz camera (board traces, three runs pooled)



H Added load: 90 Hz camera + fnn_block + dronet on the same 8 harts



I Onboard K1, measured, 45 Hz camera — CP-SAT: 57 ms, every frame on time, control 10.0 ms · greedy: 748 ms, 44/44 late, control 13.2 ms
ROS 2 8-core: 447 ms, control 16.6 ms · ROS 2 vanilla: 242 ms, control 25.6 ms

